Context-Aware Stigma Detection Model: Comparing Insider vs. Outsider Stigmatizing Language Towards People Who Use Substances

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Abstract

Online discussions about substance use are often permeated by stigmatizing language (SL), which contributes to discrimination and discourages treatment-seeking among people who use substances (PWUS). Existing stigma detection models ignore conversational context. In this study, we develop a novel, context-aware stigma detection model using a BERT-based architecture, annotator demographic information (based on jury learning), and a Deep & Cross Network v2 (DCNv2) for feature interaction. We then apply our trained model to 40,000 Reddit comments stratified across "Ingroups" and "Outgroups" [1]. To further analyze the linguistic, psychological, and thematic features that distinguish ingroup vs. outgroup and stigma positive vs. stigma negative, LIWC-based diferential language analysis (DLA) and topic modelling are conducted. By capturing the subjectivity of stigma prediction through jury learning and the implicit, conversationally dependent cases through context, our model significantly outperformed baselines (AUC=0.68 vs. AUC=0.78). We demonstrate that outsider communities use SL at significantly higher rates (84.9% prevalence), tend to associate substance use with crime, negative stereotypes, and use more third-person, negative-emotion language. In contrast, insider communities (64.1% prevalence) focus on support, recovery, and personal experience. Our findings inform the development of more effective, context-sensitive NLP models. While also elucidating the differences in linguistic elements which characterize stigma within and across cohorts. This, in turn, supports the development of less stigmatizing online communities and more effective public health strategies.

1 Introduction

Substance use, and the language surrounding it, remains a critical global public health concern. People who use substances (PWUS) regularly encounter significant social stigma, felt both in everyday interactions and throughout institutional, medical, and media discourse (Kulesza, 2014). Stigmatizing language (SL) is defined as language that communicates discrimination or negative judgment towards an identifiable group, in this case PWUS (National Institute on Drug Abuse, 2021). SL has far-reaching consequences, often undermining individuals' self-esteem, discouraging treatment-seeking, and perpetuating societal marginalization (Ashford et al., 2019). As communication about substance use increasingly migrates to digital spaces, the language on these platforms both reflects and reshapes broader attitudes toward PWUS. Despite the growing prominence of such forums, much remains unknown about the linguistic dynamics that characterize them.

Advances in computational linguistics and natural language processing (NLP) have transformed the analysis of social phenomena embedded in large-scale textual data. NLP techniques have been deployed to both predict substance use outcomes from social media language and uncover the specific linguistic features that perpetuate stigma (Jha and Singh, 2020; Giorgi et al., 2023c). Yet, much of the existing research focuses on a single piece of traditional media or broad, general-purpose platforms like Twitter or Reddit. This leaves open questions regarding how SL compares between environments, as well as, how stigma manifests in said environments.

Specifically, little is known about the difference in SL between insider communities, "ingroups" (focus on harm reduction and lived experiences) versus outsider communities, "outgroups" (primary commenters likely have no experience with substances) (See 3.2.1 for more detail). Understanding these differences is essential, as stigma in online spaces can shape community norms, influence willingness to seek support, and ultimately affect health

[1] Detailed description in Section 3.2.1

outcomes. Moreover, there is the opportunity to learn about what makes a specific community successful in being non-stigmatizing, and adopt such strategies.

Additionally, a notable limitation of most current stigma detection models is their focus on individual sentences or isolated posts, analyzed in a vacuum (Roesler et al., 2024). This neglects the broader conversational or post-level context in which SL may emerge or be interpreted. Without considering surrounding context, models risk overlooking nuanced forms of stigma or misclassifying language whose tone or meaning depends on prior discourse.

To address the above gaps, we investigate the following research questions:

(RQ1) Does incorporating conversational context in models improve the detection of stigmatizing language in online substance use discussions?

(RQ2) What are the differences between how stigma is discussed across ingroup vs. outgroup subreddits?

(RQ3) What are the key linguistic, psychological, and thematic features that distinguish stigmatizing from non-stigmatizing language within ingroups and outgroups?

To answer these questions, we employ a multi-step NLP pipeline. We sample 20,000 English-language Reddit comments from each cohort (ingroup and outgroup subreddits), enhance each with its conversational context, and apply a context-aware stigma detection model. We then conduct LIWC-based differential language analysis and topic modeling to uncover key linguistic features and thematic differences in SL across communities. This framework enables both precise stigma classification and interpretation of its linguistic and conversational nuances.

Our findings reveal both quantitative and qualitative differences in the prevalence and context of SL between ingroups and outgroups. Insider forums display lower overall rates of stigma and center topics around support, recovery, and personal experience, while outsider spaces are more likely to associate substance use with crime, law, and negative stereotypes. Additionally, our results show that incorporating conversational context will marginally improve model performance, possibly detecting more nuanced and implicit forms of stigma.

By providing a systematic, large-scale comparison of linguistic stigma dynamics across online communities, our work addresses a key gap in computational stigma research. Our findings inform both the design of context-aware NLP models and the development of more supportive, inclusive online spaces for PWUS. This research thereby advances understanding of how digital environments mediate stigma, with implications for public health and harm reduction advocacy.

Overall, this paper contributes:

1. A novel, context-aware NLP approach that enhances stigma detection performance by incorporating conversational context and annotator demographics;

2. The first large-scale quantitative and qualitative comparison of SL between insider and outsider substance-use communities, revealing linguistic and thematic patterns across cohorts.

2 Related Works

NLP methods are increasingly being used to investigate substance use, mental health, and various social science topics across media datasets.

For instance, Giorgi et al. (2023c) demonstrated that language on Twitter could predict opioid poisoning mortality per U.S. county more effectively than conventional socio-demographic and healthcare access measures, highlighting social media's strength in surfacing latent psychosocial factors.

As for stigma detection, Bouzoubaa et al. (2025) collected a dataset of 157,476 articles from The Philadelphia Inquirer (2013–2022), using BERTopic, KeyBERT, and NewSentiment to reveal that drug-related news coverage was dominated by negative and punitive language, with a consistent emphasis on crime, overdose, and law enforcement. Similarly, Giorgi et al. (2023b) conducted a linguistic analysis of New York Times articles spanning three decades, using LIWC (Linguistic Inquiry and Word Count), NRC Valence, Arousal, and Dominance lexicons, as well as word2vec embeddings, to quantify dimensions of dehumanization. They found that mainstream media frequently frames PWUS in ways that are dehumanizing. In clinical contexts, Weiner et al. (2023) analyzed 546,309 clinical notes for 30,391 patients with substance-related diagnoses using a rule-based NLP pipeline anchored to the National Institute on Drug Abuse's "Words Matter" document.

Similarly, Giorgi et al. (2023a) proposes a SL detection deep learning model which addresses the

unique challenge of subjectivity. They utilize a jury learning approach by crowdsourcing annotation of approximately 5,000 Reddit posts, with annotators' own substance use experiences recorded as key demographic features. The results demonstrate that incorporating annotator demographics into stigma detection models improves classification accuracy (AUC = 0.69) compared to language feature only baselines.

Despite all the discussed papers offering valuable insights into stigma detection models, there remains a gap in accounting for the influence of conversational and situational context. Current models that focus on standalone text may limit the models' ability to fully capture nuanced, implied, or context-dependent expressions of stigma.

As opposed to previously discussed mainstream and clinical discourse, comparing the linguistic climate of SL between environments is comparatively underexplored. Thus leaving open questions regarding the optimal communities to reduce SL for PWUS.

3 Data

3.1 Training Dataset

3.1.1 Raw Data

The annotated dataset used to develop our model was a courtesy of Giorgi et al. (2023a). In summary, 4991 unique comments were annotated by 701 MTurk workers producing 14898 total annotations. Each datapoint consists of `worker_id`, `comment_id`, `stigma_label`, `female`, `knowsud` (know someone with substance use disorder), `black`, `usesubstance` (personally use substances). The last four fields being a one hot encoding of the workers/annotators demographic information. Giorgi et al. (2023a) breakdown their methodology more in their work. Table 1 and 2 have a more granular breakdown of this dataset.

3.1.2 Preprocessing

For each comment, we enhanced contextual information by fetching the (1) `parent_text` and (2) `parent_post_text` using the Reddit API. The "parent text" is defined as the direct antecedent to a given comment: if the comment replied to another comment, the body of that comment was used. If the comment was a top-level comment on a post, the "parent text" comprised the post's title concatenated with its selftext (i.e. the body of the post). The "parent post text" is defined as the post's title concatenated with its selftext.

Statistic	Value
Total annotations	14,898
Unique comments	4,991
Unique workers	701
Avg. annotations per comment	2.99
Avg. annotations per worker	21.25

Table 1: Annotation summary statistics for the dataset.

Demographic	Count	%
(0, 1, 0, 1)	3126	20.98
(1, 1, 0, 1)	2639	17.71
(0, 0, 0, 0)	1507	10.12
(0, 0, 0, 1)	1364	9.16
(1, 1, 0, 0)	1384	9.29
(0, 1, 0, 0)	1327	8.91
(1, 0, 0, 0)	846	5.68
(0, 1, 1, 1)	742	4.98
(1, 0, 0, 1)	670	4.50
(1, 1, 1, 1)	477	3.20
(0, 0, 1, 0)	236	1.58
(0, 0, 1, 1)	172	1.15
(0, 1, 1, 0)	200	1.34
(1, 1, 1, 0)	108	0.72
(1, 0, 1, 0)	80	0.54
(1, 0, 1, 1)	20	0.13

Table 2: Annotator counts and percentages for each demographic combination, encoded as `(female, knowsud, black, usesubstance)`.

3.2 Analysis Dataset

3.2.1 Raw Data

For the purposes of this study, Reddit communities (subreddits) were divided into two distinct groups: ingroup and outgroup subreddits. The ingroup subreddits consist of r/drugs, r/addiction, and r/redditorsinrecovery, where discussions are primarily led by individuals with direct or ongoing personal experience with substance use. These communities are characterized by peer-to-peer support, harm reduction discourse, firsthand perspectives on substance use, recovery, and related challenges. In contrast, the outgroup subreddits include communities such as r/publichealth, r/worldnews, and r/advice, where the majority of participants are less likely to have direct lived experience with

Subset	# Docs	Mean Tokens/Doc	Vocab Size	K	Perplexity	c_v Coherence
Ingroup Stigma Positive	12749	52.1	3681	6	156.26	0.469
Ingroup Stigma Negative	7104	31.8	1755	10	119.49	0.506
Outgroup Stigma Positive	16968	62.9	6048	6	207.70	0.513
Outgroup Stigma Negative	3029	55.2	1399	4	109.78	0.410
Joint Cohort Stigma Positive	29171	58.0	7918	8	220.34	0.457

Table 3: Descriptive statistics and quantitative metrics for each topic model

substance use. Discussions within these subreddits may approach substance impersonally and generally reflect broader social attitudes or external perceptions. By organizing our experiments using these two groups, this study aims to examine variations in language and stigma surrounding substance use within internal communities (ingroup) versus those that engage with the topic from an external perspective (outgroup).

Reddit comments were accessed using the public archives provided by Pushshift.io (Baumgartner et al., 2020).The initial dataset included 14,447,534 comments for ingroup subreddits and 265,056,315 comments for outgroup subreddits.

3.3 Preprocessing

To focus the analysis specifically on drug-related discourse in the outgroup, we applied keyword filtering. Only comments containing at least one of the following substance-related terms were retained: "drug," "cocaine," "heroin," "lsd," "marijuana," "meth," "opiate," "opioid," "shrooms," or "weed." In contrast, the ingroup was assumed to inherently restrict discussions to be substance related due to their subreddit rules. Therefore, additional keyword filtering was deemed unnecessary for these forums. Moreover, applying excessive keyword filtering would not fully utilize the strengths of the context-aware model.

A temporal filter was also applied to both groups to limit the dataset to comments between 2015 and 2025. Leaving the ingroup with 12,147,305 comments and the outgroup with 667,253 comments.

A random subset of 20,000 comments was sampled from each group to create the final comment pool. For each sampled comment, we enhanced contextual information by retrieving the "parent text" using the Reddit API, as described in the previous training dataset section. “Parent post text” was not retrieved here as the final model used for analysis didn’t utilize it.

4 Methods

4.1 Stigma Detection Model

Our work improves the jury learning model presented by Giorgi et al. (2023a). by **(1)** extending the architecture to integrate conversational context by encoding each Reddit comment together with its immediate parent text; and **(2)** upgrading to a more expressive feature interaction scheme via a Deep & Cross Networkv2 (DCNv2) (Wang et al., 2021).

4.1.1 Pairwise Contextual Encoding

Unlike prior work, which modelled each comment in isolation our approach encodes the (comment, parent) pair within a single forward pass of BERTweet (Nguyen et al., 2020). Specifically, the input sequence is constructed as:

```
[CLS] comment_text [SEP] parent_text
[SEP]
```

However, this integration presents a unique challenge: BERTweet’s imposes a strict and unchangeable maximum sequence length of 128 tokens. Given that Reddit comments and their parents often far exceed this length, naive concatenation or off-the-shelf tokenizers are inadequate.

To address this, we implemented a manual tokenization and truncation scheme tailored to the task. With 4 special tokens automatically added by BERTweet, there was an effective budget of 124 content tokens available. We first tokenize the primary text (comment) and the secondary text (parent) separately, without adding special tokens. To prioritize the comment, we cap its representation at 96 tokens or less, according to availability. The remaining token budget of at least 28 is then allocated to the parent text.This formulation enables the model to attend jointly across comment and parent, allowing genuine cross-utterance interactions to be modeled in the text encoder. The BERTweet backbone is kept frozen throughout training, and

the pooled representation is obtained via attention-masked mean pooling.

The resulting 768-dimensional text embedding is concatenated with a four-dimensional binary demographic vector (female, knowsud, black, usesubstance) and an eight-dimensional learned embedding which identifies the specific worker. This 780-dimensional input vector captures, for each annotation, both the linguistic and social context as well as the identity of the annotator.

4.1.2 Deep & Cross Network v2 (DCNv2) Classifier

To facilitate both explicit and implicit feature interactions, we employ a DCNv2 (Wang et al., 2021). The cross network consists of three vector-valued cross layers, each parameterized by a full-rank matrix, enabling higher-capacity modeling of cross-feature effects than previous rank-1 DCN layers. Parallel to this, a two-layer feed-forward deep network captures non-linearities. The outputs of the cross and deep subnetworks are concatenated and passed through a final linear layer to predict the stigma label for each (comment, annotator) pair. All classifier parameters, including annotator embeddings, are trained jointly.

4.1.3 Training

The model is trained with binary cross-entropy loss using the AdamW optimizer with a learning rate of 1×10^{-3} and weight decay 1×10^{-4}. An 85/15 test-train split at the comment-level is used to develop the model. Whereas averaging over a 5-fold cross validation is used to report the final performance metrics.

4.1.4 Inference

To address RQ2 and RQ3, the model must be used to detect stigma in unseen data. A jury prediction approach is adopted to model the subjectivity of stigma detection across demographic profiles (Gordon et al., 2022). We created a jury with 14 jurors by using the binary demographic vector (female, knowsud, black, usesubstance). Note that there are $2^4 = 16$ unique combinations for this four-dimensional vector, however (1,0,1,0) and (1,0,1,1) were dropped due to minimal representation in the training set (See Table 2), thus producing 14 jurors. For each comment, a prediction is generated from each simulated juror, and the final stigma label is determined by majority vote and a well-defined tie-breaker policy.

As for the other two model inputs, the input text embedding is generated using the same preprocessing and pairwise tokenization strategy as in training. On the other hand, the annotator embedding is usually chosen from the worker id field in the training data, which doesn't exist for new data. The strategy we employ is to simply find the mean of all learned annotator embeddings and to use that vector as the only annotator embedding. Essentially telling the model to treat every annotator as the average annotator. This approach puts the focus on demographic subjectivity as opposed to annotator variance.

4.2 Differential Language Analysis (DLA)

DLA was conducted on both the ingroup and outgroup. Predictors consisted of the full Linguistic Inquiry and Word Count (LIWC) lexicon (LIWC). Each feature represents the relative frequency of that category's tokens within a comment. After discarding categories absent from the processed dataframe, 77 LIWC features were retained for modeling.

To stabilize maximum-likelihood estimation and to render coefficients comparable across categories of differing base rates, all 77 predictors were z-score standardized prior to model fitting. The association between LIWC features and the stigma label was estimated with a L2-regularized binary logistic regression.

4.3 Topic Model

To investigate thematic patterns in both stigmatizing and non-stigmatizing discourse, we applied topic modeling using Latent Dirichlet Allocation (LDA) (Jelodar et al., 2018) separately to ingroup and outgroup comments, further stratified by stigma label (positive or negative). This yielded four distinct topic models, each capturing dominant themes specific to their subgroup.

For each subset, documents were first preprocessed by lowercasing texts, stripping URLS and HTML entities, and removing all non-alphabetic characters. Common stopwords, as well as custom Reddit- and chat-specific filler words, were filtered out. Lemmatization was employed to reduce words to their base forms (Loper and Bird, 2002). Additionally, frequent bigrams were detected and consolidated using the gensim library (Rek and Sojka, 2010), and only tokens with a minimum length of three characters were retained.

A dictionary and bag-of-words (BoW) corpus were constructed from the cleaned texts. The final

Model	F1	Accuracy	AUC
Baselines			
LIWC Logistic Regression	0.55	0.54	0.59
Jury Learning	0.68	0.68	0.68
Experiments			
Jury Learning (DCNv2)	0.72	0.69	0.76
Jury Learning (DCNv2) + parent_text	**0.72**	**0.70**	**0.78**
Jury Learning (DCNv2) + parent_text + parent_post_text	0.72	0.69	0.78

Table 4: Performance metrics for model development experiments

vocabulary was trimmed by filtering out tokens that appear in less than 20 documents or more than 50% of documents (`no_below=20, no_above=0.5`). The number of topics (K) for each model was selected by sweeping K values from 4 to 20 and choosing the value with the highest coherence score (c_v). Final LDA models were then trained using optimized hyperparameters and the selected number of topics. Key corpus and model statistics are summarized in Table 3. All models produced interpretable topics (c_v > 0.50) and achieved reasonable perplexity.

4.4 Joint Cohort, Stigma Positive Topic Model

To enable direct comparison of thematic content between insider and outsider communities within stigma-positive discourse, we conducted a joint topic modeling analysis. Contrary to the individual models described above, a single LDA model was trained on the combined corpus of stigma-positive comments drawn from both ingroup and outgroup cohorts. Preprocessing steps followed the same pipeline as in our individual topic modeling. Training metrics are in Table 3.

The model was used to infer topic distributions for every document in the dataset. For each document d, the model outputs a probability vector indicating the assigned weight for each topic,

$$[P(T_0|d), P(T_1|d), \ldots, P(T_{K-1}|d)]$$

. These were aggregated into a document-topic matrix, with each row representing a document and each column a particular topic.

Each document was tagged with its original cohort label (ingroup or outgroup). This enabled separation of the full set of documents into the two groups. We then computed the soft prevalence of each topic within both cohorts. For topic k, the soft prevalence in a cohort was calculated as the mean topic probability across all documents from that cohort:

$$P(k|\text{drug}) = \frac{1}{N_{\text{ingroup}}} \sum_{d \in \text{ingroup}} P(T_k|d)$$

$$P(k|\text{outgroup}) = \frac{1}{N_{\text{outgroup}}} \sum_{d \in \text{outgroup}} P(T_k|d)$$

where N_{drug} and N_{outgroup} denote the number of documents in the ingroup and outgroup, respectively.

Finally, for each topic, the cohort difference was computed as:

$$\Delta_k = P(k|\text{ingroup}) - P(k|\text{outgroup})$$

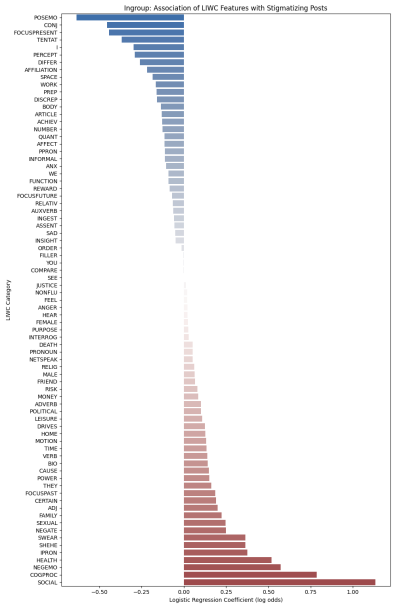
which quantifies the relative prominence of topic k in the insider compared to the outsider cohort. This approach ensures topic representations are directly comparable across cohorts, as all documents are projected into a shared latent topic space.

We assessed whether the prevalence of each topic differed significantly between comments from ingroup and outgroup using two complementary statistical tests. First, we applied a two-proportion z-test to compare the proportion of comments in each group where a given topic was the most prevalent ("hard" topic assignment). Second, we used Welch's t-test to compare the average topic proportions ("soft" assignments) between groups. All p-values were corrected for multiple comparisons using the Benjamini–Hochberg procedure. For the two-proportion tests, we also report Cohen's h as a measure of effect size.

5 Results

5.1 Context-Aware Model

Table 4 shows the results of the model development experiments. The metrics were determined



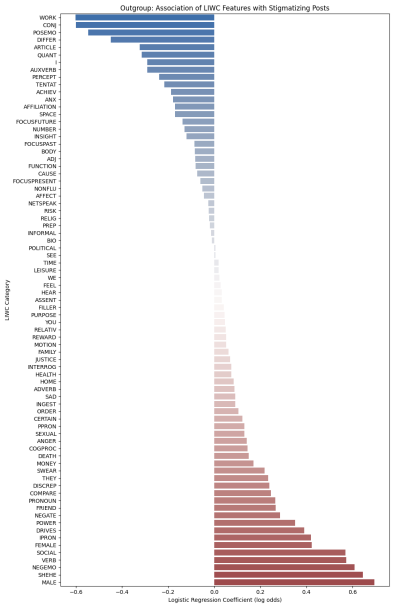
(a)

Feature	Definition
POSEMO	Positive emotion
CONJ	Conjunctions
FOCUSPRESENT	Present-focus
TENTAT	Tentativeness
I	First person singular (*I*)

(b)

Feature	Definition
SOCIAL	Social processes
COGPROC	Cognitive processes
NEGEMO	Negative emotion
HEALTH	Medical context
IPRON	Impersonal pronouns

(c)



(d)

Feature	Definition
WORK	Work-related words
CONJ	Conjunctions
POSEMO	Positive emotion
DIFFER	Differentiation
ARTICLE	Articles (*a, an, the*)

(e)

Feature	Definition
MALE	Male references
SHEHE	Third-person pronouns (she/he)
NEGEMO	Negative emotion
SOCIAL	Social processes
VERB	Verbs

(f)

Figure 1: DLA results showing association of LIWC features with SL in ingroup (left) and outgroup (right). Bar plots (a) and (d) show log odds for every existing LIWC feature. (b) and (e) explicitly state and define the top 5 stigma negative LIWC features for their respective groups. (c) and (f) do the same but for the stigma positive features

through averaging scores across a 5-fold cross validation. We can see that the move to a DCNv2 classifier resulted in the most substantial performance gains. Parent text addition marginally outperformed DCNv2 baseline. Next, we experimented including a given comment's parent post as additional context, however this did not perform better. We hypothesize that the performance gains given by context addition could be limited by how the training dataset was obtained by Giorgi et al. (2023a). In that, comments were selected through keyword matching with explicit substance related terms. This minimizes the representation of implicit stigma and doesn't account for comments which have no substance keyword, but reply to substance related comments/posts—the exact situations where a context-aware model would thrive.

Nonetheless, context addition still resulted in notable gains to accuracy and AUC. As the best-performing model, the DCNv2 + parent text model was used for downstream analyses.

5.2 Stigma Detection and Prevalence

We employed our stigma detection model to predict the prevalence of SL in both ingroup and outgroup samples (each consisting of 20,000 comments). As presented in Table Y, the model identified SL in 64.1% of ingroup comments (n = 12,828) and

84.9% of outgroup comments (n = 16,970), indicating a substantially higher prevalence of stigma in the outgroup cohort.

5.3 DLA

To systematically visualize the linguistic elements associated with stigma-positive versus stigma-negative comments, DLA was conducted on both the ingroup and outgroup using LIWC features (Figure 1).

5.4 Topic Model

Topic modelling across ingroup and outgroup (further stratified by stigma label) is visualized by the word clouds in Figure 2. Only the top four most prevalent topics for each group are presented. Each topic visualization was constrained to 20 words. Model performance metrics are summarized in Table 3.

Several observations can be made here comparing across stigma labels and/or subreddit cohorts. For readability, the notation (group, topic number) will be used here. For example, (IS+, 2) refers to "Ingroup Stigma Positive Topic 2".

5.4.1 Cross-Label Comparison: Ingroup

Firstly, (IS+,0) and (IS-, 0) largely overlap, suggesting this form of language is common in the ingroup (not a strong indicator of stigma). This can be similarly applied to (IS+, 4) and (IS-, 4). On the other hand, (IS+, 2) suggests there is a correlation between stigma, talking about others ("friend"), and cuss words. As for associations with the absence of stigma, the most prevalent topics are notably experience-focused. For example, (IS-, 8) is dominated by more informal vocabulary/slang ("trip", "lsd", "acid", "shrooms"), hinting at a potentially more personal and friendly communication style. (IS-, 6) centers on temporal and behavioral terms like "day", "week, "month" in the context of patterns of usage, quitting, and personal routines (e.g., "stop," "feel," "quit," "sober").

5.4.2 Cross-Label Comparison: Outgroup

In the outgroup, stigma-positive comments largely have associations with crime, war, government or the cartel as shown by (OS+, 0) and (IS+, 2). Whereas a stigma-negative label seems to correlate with the research (OS-, 3) and economic (OS-, 0) side of drug-related discourse.

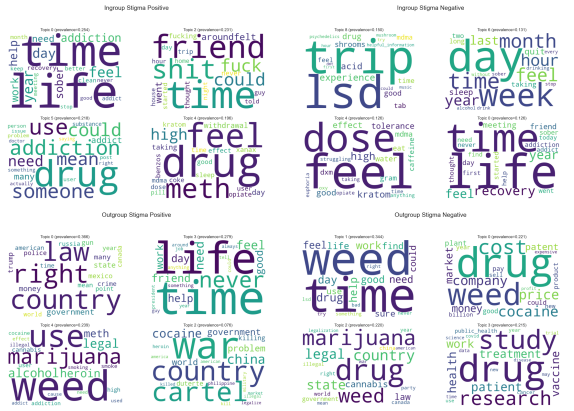


Figure 2: Word clouds generated from a LDA topic model for each cohort and stigma label.

5.5 Joint Topic Model: Cross-Cohort SL Comparison

Figure 3 elucidates more of how SL differs between ingroup and outgroup. All 8 out of 8 topics showed significant differences in prevalence between the ingroup and outgroup cohorts after Benjamini–Hochberg FDR correction ($q < 0.05$, two-proportion z-test). Results were also consistent with Welch's t-test on average topic proportions. Effect size was also measured to produce a Cohen's h range from 0.11–0.57.

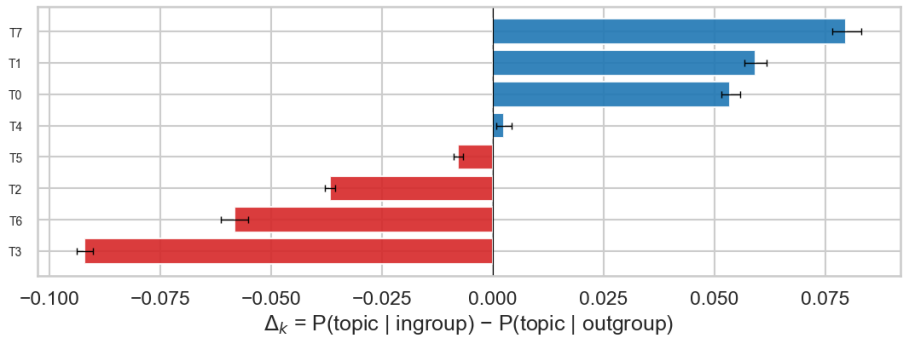
Amongst the topics overrepresented in ingroup subreddits, T7 suggests references to personal life, feelings, and experiences. T1 focuses on slang and casual language. T0 has references to medical contexts and effects.

As for outgroup skewed topics, T2 is centralized on economic and legal discussions. T6 features more depersonalized or "problem-focused" language. T3 clearly demonstrates the outgroup's tendency to associate drug use with societal/criminal issues

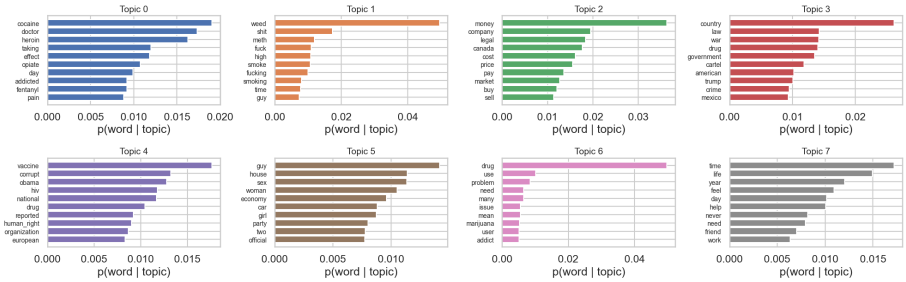
6 Discussion

6.1 Stigma Detection Model

The modelling results contribute to the NLP side of the literature. First, building off of Giorgi et al. (2023a) by replacing the original classifier with DCNv2 yielded the largest gain overall. This supports the argument made by Giorgi et al. (2023a) that annotator characteristics matter in stigma detection, and further suggests that richer feature interaction architectures can better capture how linguistic signals and social interpretations combine. Second, the addition of parent-text context yielded



(a) Differential topic prevalence (stigma-positive cross cohort). Blue=ingroup skewed; red=outgroup skewed. Bootstrap confidence intervals are included.



(b) Top 10 words per topic

Figure 3: Joint topic model experiment combined ingroup and outgroup, stigma positive documents.

a modest but consistent improvement in accuracy and AUC, even though the gain was smaller than the effect of the classifier upgrade itself. This finding is still theoretically important. It indicates that stigma is not always contained within a single isolated comment; at least some instances depend on the immediately preceding conversational context for interpretation.

We stipulate that the relatively modest improvement from context likely reflects the way the training data was originally constructed. Because the annotation dataset was keyword-selected around explicit substance terms, it may underrepresent precisely the kinds of implicit, referential, or conversationally dependent cases where context would matter most. A context-aware model trained on thread-level data that includes indirect references, stance shifts, sarcasm, and replies without explicit drug keywords would likely show larger gains, and would be an interesting future work.

The lack of further improvement from adding parent_post_text may indicate that local reply structure is more useful than broader thread context, or simply reflect the constraints of the training data and token budget.

6.2 Analysis

Applying our stigma detection model to external data revealed that outsider-oriented communities exhibited substantially higher rates of predicted stigma than insider-oriented communities. Outgroup stigma was more likely to frame substance use through crime, conflict, policy, and social disorder, whereas ingroup discourse was more often grounded in personal experience, everyday life, bodily effects, recovery, and peer-to-peer interaction. This pattern directly supports the broader premise of this study: online stigma toward PWUS is context-sensitive, and the discourse of communities closer to lived experience differs meaningfully from that of more socially distant communities.

These results align with prior work showing that mainstream and institutional discourse often represents substance use through punitive, dehumanizing, or problem-focused frames. Bouzoubaa et al. (2025) found that drug-related news coverage was dominated by negative language emphasizing crime, overdose, and law enforcement, while Giorgi et al. (2023b) showed that major newspaper coverage frequently dehumanized PWUS. Our outgroup results suggest that such framing is not limited to formal media institutions: similar pat-

terns appear in everyday online discussion as well. The outgroup topic models and joint topic analysis repeatedly surfaced themes centered on law, cartels, war, government, economics, and generalized "drug problems," indicating that outsider communities often discuss substance use at the level of public threat rather than lived human experience. In this sense, our findings extend previous media-focused work by showing that stigmatizing narratives are reproduced not only by journalists or institutions, but perhaps is characteristic of an "outsider-voice" speaking about PWUS. Notably, some outgroup stigma-negative topics were associated with research, treatment, and economic or policy discussion, which suggests that outsider discourse is not inherently stigmatizing.

Psychology research suggests that stigmatized groups are often discussed in ways that deny full humanity, agency, or uniquely human emotional complexity (Leyens et al., 2007). This provides a useful lens for interpreting our outgroup DLA results: stigma-positive outsider discourse was associated with more third-person, male-referential, and negative-emotion language. Rather than reflecting only topic differences, these patterns are consistent with a more socially distant and categorical style of talking about PWUS (Sumnall et al., 2021).

By contrast, in the DLA, stigma-negative ingroup comments were associated with positive emotion, present-focus, tentativeness, and first-person singular language. The topic models reinforce this interpretation, with non-stigmatizing ingroup topics organized around personal routines, recovery timelines, dosing, and informal community language. This pattern aligns with work on online recovery and harm-reduction communities showing that peer support spaces are often organized around mutual recognition, practical knowledge exchange, and experiential expertise (Balsamo et al., 2023; Engel et al., 2021).

However, our results also show that insider communities should not be treated as stigma-free spaces. The model still identified SL in a substantial proportion of ingroup comments, even if the prevalence was markedly lower than in the outgroup. Rather than disappearing, stigma appears to change form. In the ingroup, stigma-positive language was less dominated by criminalization or broad societal threat narratives. The DLA results show that ingroup stigma-positive comments were associated with social processes, cognitive processes, negative emotion, health language, and impersonal pronouns. The topic models similarly suggest that stigmatizing ingroup discourse can emerge when users talk about other people ("friend," "someone"), addiction labels, withdrawal, or more negatively charged substance categories. One plausible interpretation is that insider spaces contain forms of internalized stigma (Roesler et al., 2024).

Finally, we highlight the benefit of "insider" communities by showing the discussions and linguistic elements which result in significantly lower SL prevalence. Our findings resonate with (Engel et al., 2021), in that they argue for the benefit of the harm reduction community, Bluelight.org, for PWUS.

7 Conclusion

This study demonstrates the power of context-aware NLP models for detecting and characterizing SL in online substance use discussions. By integrating conversational context and annotator demographics, our approach outperforms traditional context-free baselines, highlighting the need to account for both social and linguistic dimensions in stigma research. Our large-scale analysis reveals that outsider communities engage in stigmatizing discourse at significantly higher rates, often focusing on crime and negative stereotypes, while insider communities, though not stigma-free, emphasize support and personal experience. DLA and topic modeling further illuminate how specific linguistic patterns distinguish stigma across community boundaries, as well as, how SL manifests within each community. These findings underscore the importance of community context in understanding and combating online stigma, with direct implications for public health messaging, harm reduction advocacy, and automated moderation strategies.

Limitations

Despite providing novel insights into the linguistic dynamics of SL across online platforms, our study is subject to several limitations. First, data collection was constrained by access to platform-specific content. While Bluelight.org and Reddit offer rich sources of discourse relevant to harm reduction and substance use, the findings of our analysis may not generalize to other online communities, non-digital spaces, or broader populations of PWUS. Additionally, our dataset consists only of posts in English,

which restricts our analysis to English-speaking users and may overlook important cross-cultural differences in language and stigma. Future research should address these limitations by expanding to a broader range of platforms, including non-English language communities.

A further limitation of this research stems from the fact that the stigma detection model was trained exclusively on Reddit data. Each online platform has its own unique linguistic conventions, community norms, and styles of communication, which may not be fully captured by a model tuned to a single source. As a result, the model may fail to recognize, or may misinterpret, examples of stigmatizing language unique to other platforms such as Bluelight. This could potentially lead to under- or over-estimation of stigma in contexts where language use diverges from Reddit-specific patterns. Future work could address this limitation by adapting and validating stigma detection models on platform-specific datasets, allowing for more robust and generalizable analyses across diverse online communities.

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